WORKSHEET

Chapter 12: Solving simultaneous equations

1 Use the substitution method to solve the following pairs of simultaneous equations:

a $y = 1 + x$ and $y = 8$

b $y = 1 - x$ and $y = -2$

c $y = 1 + x$ and $y = 1 - x$

d $y = -1 + x$ and $y = 1 - x$

e $y = 3x$ and $y = 8 - x$

f $y = -2x$ and $y = x$

g $y = 3 + x$ and $y = 3 - 2x$

h $y = 2 + 2x$ and $y = x$

i $x = y + 1$ and $x = 2y - 1$

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j $x = 3y$ and $x = 4 - y$

k $y = \frac{1}{2}x$ and $y = 6 + x$

l $y = 5 + 2x$ and $y = -3x$

m $y = 1 + 2x$ and $y = 2 - x$

n $y = 4 - x$ and $y = -1 + 2x$

o $y = -2 + x$ and $y = 1 - 3x$

2 Use the elimination method to solve the following pairs of simultaneous equations:

a $x + y = 1$ and $x - y = 1$

b $y = 4 - x$ and $y = 2 + x$

c $2x + y = 1$ and $x - y = 5$

d $3y = 5 - x$ and $y = 3 + x$

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e $y - 2x = 3$ and $-y + x = 1$

f $y = 2 - 3x$ and $2y = 1 + 3x$

g $2y - 3x = 2$ and $y + 3x = 4$

h $4y = 3 + 2x$ and $4y = 1 + x$

i $3y + 2x = 7$ and $-3y + x = 2$

j $5y - 4x = 4$ and $2y + 4x = 3$

k $x = 3y - 4$ and $-x = 2y - 1$

l $x - 2y = 7$ and $3x + 2y = 1$

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m $3y = 8 - x$ and $2y = 7 + x$

n $4x = 2 + 3y$ and $4x = 4 + y$

o $x + 8y - 1 = 0$ and $2x - 8y + 7 = 0$

3 Using either method, solve the following pairs of simultaneous equations:

a $y = 5 + x$ and $y = -1 + 2x$

b $x + y = 1$ and $x - y = 7$

c $y = 6 - x$ and $y = 2 + 3x$

d $2y = 1 + 2x$ and $y = 2 - 2x$

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e $y = 3 + x$ and $y = -2 + 2x$

f $2y - x = 3$ and $3y + x = 2$

g $y + 3x = 0$ and $2y - 3x = 6$

h $y = 6 + 5x$ and $y = -2 + 3x$

i $y = 3 - 2x$ and $y = 2 + 3x$

j $2y = 4 + x$ and $-2y = 3 + 2x$

k $x = 5y$ and $x = 2y - 9$

l $x + y - 6 = 0$ and $x - y - 2 = 0$

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m $y = 2 - 5x$ and $y = -12 + 2x$

n $x = 2y$ and $x = y - 5$

o $2x + y - 7 = 0$ and $-2x + y + 3 = 0$

p $3y = 4 - x$ and $-y = 4 + x$

q $y = 3 - x$ and $y = 2 + x$

r $x + 3y - 6 = 0$ and $2x - 3y - 3 = 0$

s $2x = 4y - 1$ and $3x = -4y + 6$

t $y = 8 - 3x$ and $y = 5 - 2x$

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Answers

1 **a** $(7, 8)$ **b** $(3, -2)$ **c** $(0, 1)$
d $(1, 0)$ **e** $(2, 6)$ **f** $(0, 0)$
g $(0, 3)$ **h** $(-2, -2)$ **i** $(3, 2)$
j $(3, 1)$ **k** $(-12, -6)$ **l** $(-1, 3)$
m $\left(\frac{1}{3}, 1\frac{2}{3}\right)$ **n** $\left(1\frac{2}{3}, 2\frac{1}{3}\right)$ **o** $\left(\frac{3}{4}, -\frac{5}{4}\right)$

2 **a** $(1, 0)$ **b** $(1, 3)$ **c** $(2, -3)$
d $(-1, 2)$ **e** $(-4, -5)$ **f** $\left(\frac{1}{3}, 1\right)$
g $\left(\frac{2}{3}, 2\right)$ **h** $\left(-2, -\frac{1}{4}\right)$ **i** $\left(3, \frac{1}{3}\right)$
j $\left(\frac{1}{4}, 1\right)$ **k** $(-1, 1)$ **l** $\left(2, -2\frac{1}{2}\right)$
m $(-1, 3)$ **n** $\left(1\frac{1}{4}, 1\right)$ **o** $\left(-2, \frac{3}{8}\right)$

3 **a** $(6, 11)$ **b** $(4, -3)$ **c** $(1, 5)$ **d** $\left(\frac{1}{2}, 1\right)$
e $(5, 8)$ **f** $(-1, 1)$ **g** $\left(-\frac{2}{3}, 2\right)$ **h** $(-4, -14)$
i $\left(\frac{1}{5}, 2\frac{3}{5}\right)$ **j** $\left(-2\frac{1}{3}, \frac{5}{6}\right)$ **k** $(-15, -3)$ **l** $(4, 2)$
m $(2, -8)$ **n** $(-10, -5)$ **o** $\left(2\frac{1}{2}, 2\right)$ **p** $(-8, 4)$
q $\left(\frac{1}{2}, 2\frac{1}{2}\right)$ **r** $(3, 1)$ **s** $\left(1, \frac{3}{4}\right)$ **t** $(3, -1)$

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